HACKATHON 2024

Your Buddy

Your AI-Powered Study Companion

An intelligent RAG system that transforms documents, videos, and textbooks into quizzes, summaries, and instant answers — powered by Google Gemini.

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The Problem: Information Overload

📚 Dense Content

Students drown in PDFs, lecture notes, and videos with no smart way to extract key insights.

⏱️ No Time to Review

Manual summarization and quiz creation is time-consuming and inefficient before exams.

🔍 Static Search

Traditional search can't answer questions grounded in *your* specific uploaded content.

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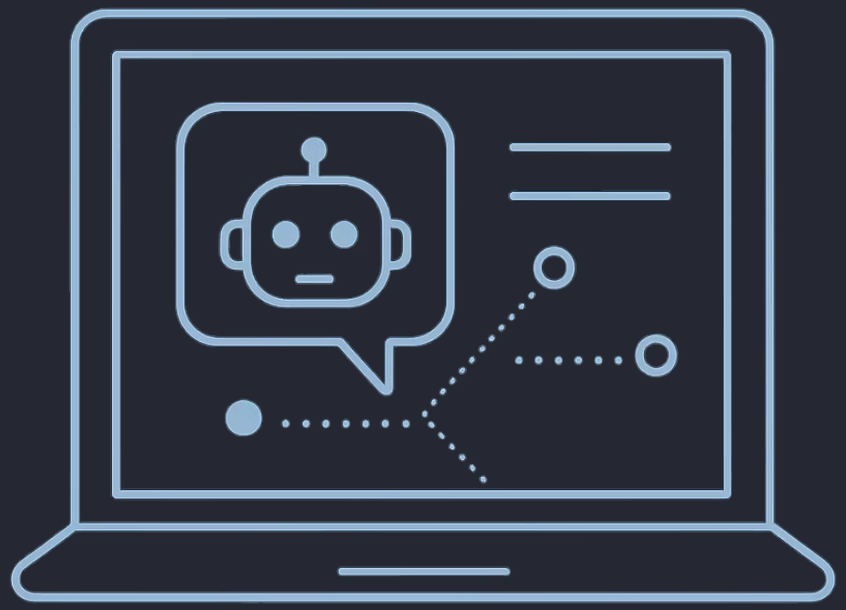
Introducing Your Buddy



Your Buddy is an AI-powered Retrieval-Augmented Generation (RAG) system — inspired by NotebookLM — that lets you upload documents, ask questions, and generate intelligent study materials in seconds.

- Upload any PDF
- Chat with your content
- Generate quizzes & summaries

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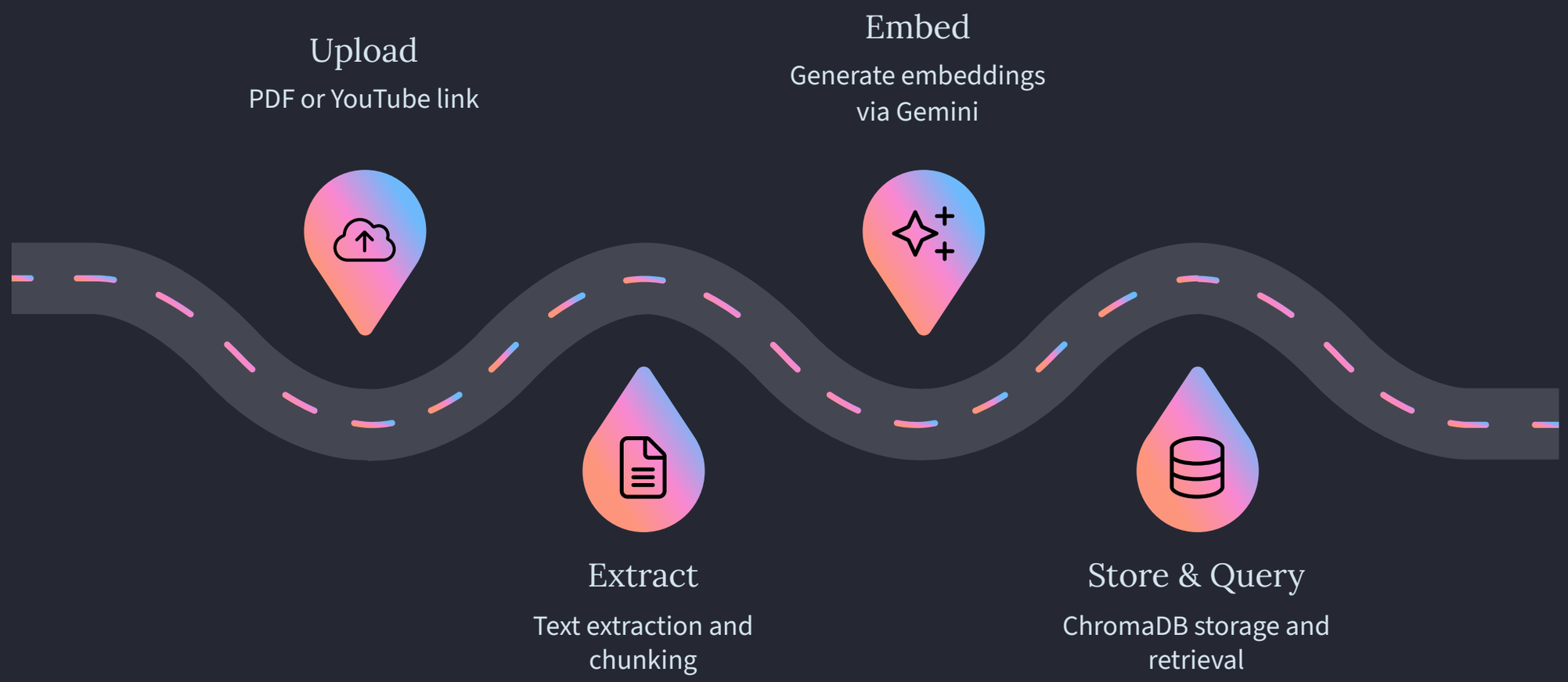
System Architecture

Your Buddy's end-to-end pipeline connects user input to AI-powered output through a structured RAG flow — from document ingestion to intelligent response generation.



Every query is grounded in your uploaded content — no hallucinations, just precise, sourced answers powered by Google Gemini.

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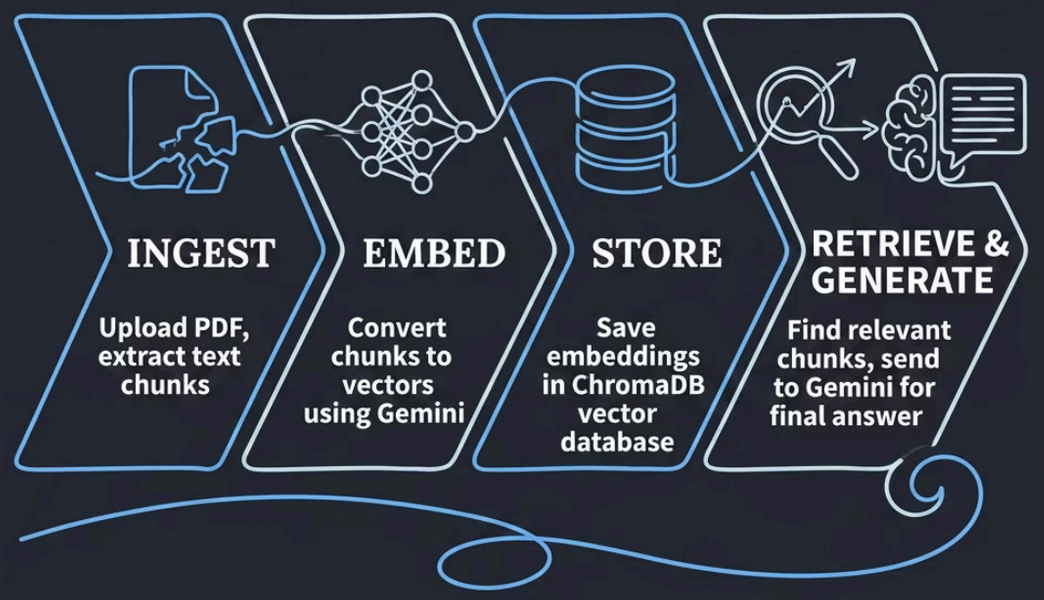
How RAG Works Here

Retrieval-Augmented Generation combines the precision of vector search with the fluency of large language models. Your Buddy embeds your documents, stores them in ChromaDB, and retrieves the most relevant chunks before generating an answer.

> ⓘ Unlike standard LLMs, RAG grounds every response in *your* actual content — making answers accurate, relevant, and citation-ready.



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Core Features

PDF Upload & Knowledge Extraction

Upload any PDF and instantly extract structured, searchable knowledge.

AI-Generated Quizzes

Select any chapter or section and auto-generate targeted study quizzes.

Gemini-Powered Chatbot

Ask questions about your content and get accurate, sourced answers.

YouTube Summarizer

Paste a video link and extract key insights and summaries automatically.

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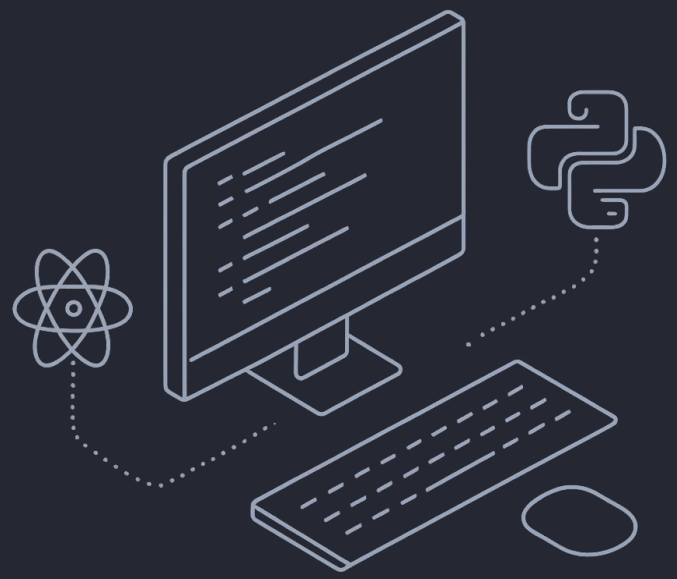
Tech Stack



⚛️ React

Responsive, component-driven frontend deployed on Firebase Hosting.

🐍 FastAPI (Python)

Lightweight, async backend API handling all AI and data logic.

🗄️ ChromaDB

Lightweight vector database for fast semantic similarity search.

✨ Google Gemini API

Multimodal AI model powering embeddings, chat, and content generation.

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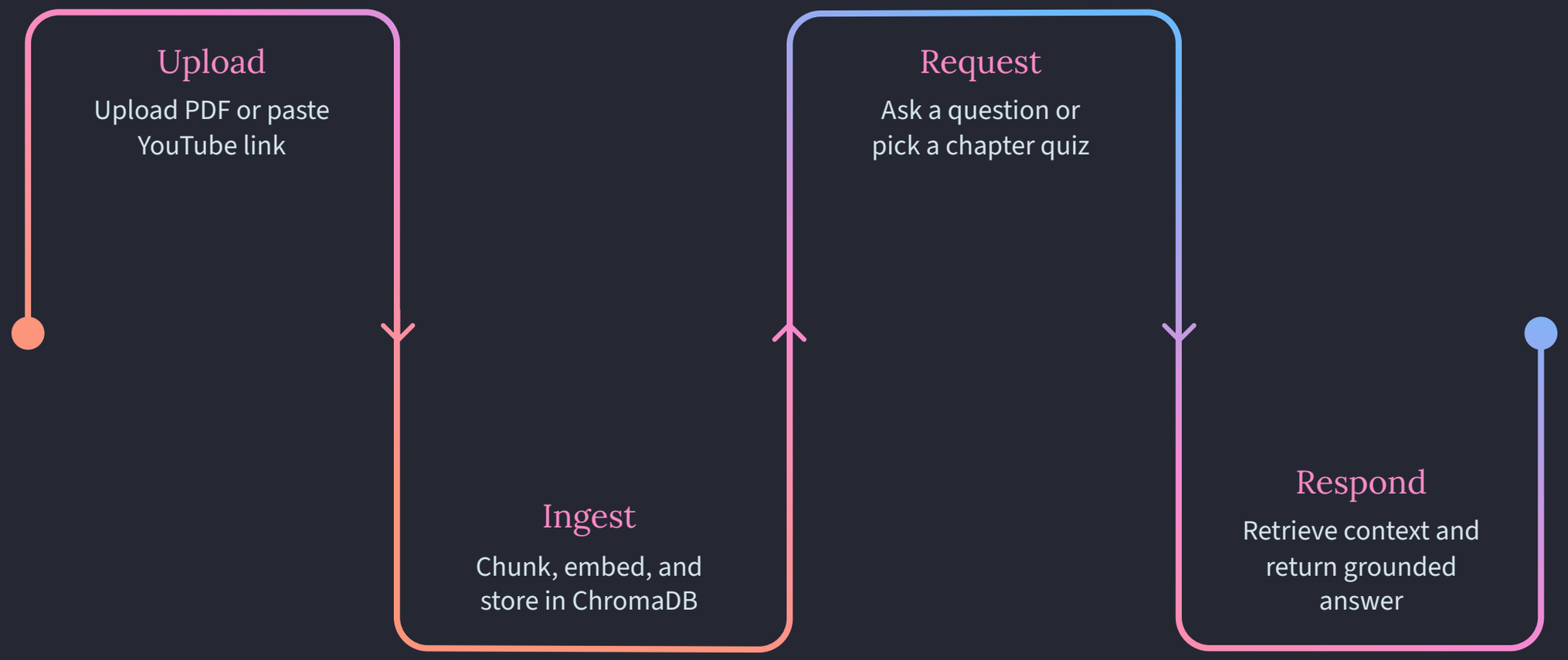
Demo Workflow

From upload to insight in under 60 seconds — here's the complete user journey.



The entire pipeline — from document ingestion to intelligent output — runs seamlessly in real time, giving students instant study materials.

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Challenges We Overcame

1

Chunking Strategy

Finding the right text chunk size to balance context quality with retrieval accuracy required careful experimentation.

2

PDF Parsing Variability

Handling diverse PDF formats — columns, tables, and scanned pages — demanded robust preprocessing logic.

3

Latency Optimization

Minimizing response time for embedding generation and vector search while maintaining answer quality.

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What's Next for Your Buddy

🔗 Multi-Source Support

Support URLs, DOCX, and image-based documents.

👤 User Accounts

Persistent libraries, history, and personalized study plans.

📊 Analytics Dashboard

Track study progress, quiz scores, and content coverage.

🌐 Mobile App

Native iOS and Android apps for on-the-go studying.

Thank You

Your Buddy redefines how students interact with knowledge — turning passive reading into active, AI-assisted learning.

🔗 **GitHub:** github.com/4chuck/your_buddy

REACT | FASTAPI | CHROMADB | GEMINI AI

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